WTW 158 2016 Proofs to be studied for the exam:

1. Differentiability implies continuity

Theorem
If a function f is differentiable at a then f is continuous at a.

(So the theorem says if $\lim\limits_{x\to a} \frac{f(x)-f(a)}{x-a} = f'(a)$ (a finite number) then $\lim_{x\to a} f(x) = f(a)$)

Proof
$\lim\limits_{x\to a} f(x)$
$= \lim\limits_{x\to a}(f(x) - f(a) + f(a))$ Addition or subtraction of the same term
$= \lim\limits_{x\to a}(f(x) - f(a)) + \lim\limits_{x\to a} f(a)$ Limit law of addition
$= \lim\limits_{x\to a} \frac{f(x)-f(a)}{x-a}(x-a) + f(a)$ Multiplication and division by the same term
$= \lim\limits_{x\to a} \frac{f(x)-f(a)}{x-a} \cdot \lim\limits_{x\to a}(x-a) + f(a)$ Limit law of multiplication
$= f'(a) \cdot 0 + f(a)$ f is differentiable
$= f(a)$

So $\lim\limits_{x\to a} f(x) = f(a)$ and therefore f is continuous in $x = a$.

Note: If f is continuous in a then f is not necessarily differentiable in a.
Counter example: $f(x) = |x|$ is continuous in 0 but not differentiable in 0.

2. (i) A positive derivative indicates that the function is increasing.

Definition of an increasing function:
A function f is increasing on an interval I if $f(x_2) > f(x_1)$ for any $x_1, x_2 \in I$ such that $x_2 > x_1$.

Theorem: If $f' > 0$ on an interval I then f is increasing on the interval.

Proof: Pick any two points $x_1, x_2 \in I$ such that $x_2 > x_1$.
Consider the function f on the interval $[x_1, x_2]$.

1. f is continuous on $[x_1, x_2]$ (f is differentiable on I)
2. f is differentiable on (x_1, x_2) (f is differentiable on I)

So, according to the Mean Value Theorem a $c \in (x_1, x_2)$ exists such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

But $f'(c) > 0$ and $x_2 - x_1 > 0$, therefore $f(x_2) - f(x_1) > 0$.
It follows that $f(x_2) > f(x_1)$ and therefore f is increasing on I.

2 (ii) A negative derivative indicates that the function is decreasing.

Definition of an decreasing function:
A function f is decreasing on an interval I if $f(x_2) < f(x_1)$ for any $x_1, x_2 \in I$ such that $x_2 > x_1$.

Theorem: If $f' < 0$ on an interval I then f is decreasing on the interval.

The proof of this theorem is similar to that given above.

3. The Fundamental Theorem of calculus

Theorem

If f in continuous on $[a,b]$ and if F is any anti-derivative of f, that is $F' = f$, then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Proof

Divide the interval $[a,b]$ in n subintervals with equal length $\triangle x = \dfrac{b-a}{n}$.

Let $x_0 = a < x_1 < \ldots < x_{n-1} < x_n = b$ be the endpoints of the subintervals. ....(**A**)

Let F be an anti-derivative of f, that is F is a function such that $F' = f$. (**B**)

We apply the Mean Value Theorem to F on $[x_{i-1}, x_i]$:

- F is differentiable on $[a,b]$ (if $F' = f$, then the derivative exists) so F is differentiable on $[x_{i-1}, x_i]$
- F is continuous on $[a,b]$ (because F is differentiable). so F is continuous on $[x_{i-1}, x_i]$

So there existes a number $c_i \in (x_{i-1}, x_i)$ such that

$$\frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} = F'(c_i)$$

$$= f(c_i) \qquad \text{(from B)} \qquad (\mathbf{C})$$

Take the point c_i as the sample point in every interval.

Because f is continuous on $[a,b]$, then f is integrable on $[a,b]$. It follows that

$$\int_a^b f(x)\,dx$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\Delta x$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}}\Delta x \qquad \text{From the Mean Value Theorem (} \mathbf{C} \text{ above)}$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}}(x_i - x_{i-1})$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} F(x_i) - F(x_{i-1})$$

$$= \lim_{n\to\infty} \big([F(x_1) - F(x_0)] + [F(x_2) - F(x_1)] + [F(x_3) - F(x_2)] + \ldots + [F(x_{n-1}) - F(x_{n-2})] + [F(x_n) - F(x_{n-1})]$$

$$= \lim_{n\to\infty} \big([F(x_n) - F(x_0)]$$

$$= \lim_{n\to\infty} \big([F(b) - F(a)] \qquad \text{(From } (\mathbf{A})\text{)}$$

$$= F(b) - F(a)$$